

Your grade: **96%**

Your latest: **96%** • Your highest: **96%** • To pass you need at least 50%. We keep your highest score.

Next item →

1. When saving a Keras model, which of the following is NOT typically saved by model.save?

1 / 1 point

- ☐ Optimizer state
- ☒ Dataset used for training
- ☐ Training configuration
- ☐ Model architecture

 **Correct**

Correct! The dataset used for training is not saved by model.save.

2. What are the benefits of saving a model in Keras?

1 / 1 point

- ☐ Makes the model architecture readable
- ☒ Allows for model reuse without retraining

 **Correct**

Correct! Saving a model allows it to be reused without needing to retrain it.

- ☒ Ensures reproducibility of results

 **Correct**

Correct! Saving the model ensures that the results can be reproduced.

- ☒ Facilitates model versioning and tracking

 **Correct**

Correct! Saving models helps in versioning and tracking different model iterations.

- ☐ Automatically optimizes model performance

3. When interpreting prediction results, what does a predicted class with a high probability score indicate?

1 / 1 point

- ☐ The input data is not suitable for the model
- ☐ The model is unsure and the result is likely wrong
- ☐ The model needs retraining for better accuracy
- ☒ The model is confident in its prediction for that class

 **Correct**

Correct! A high probability score indicates that the model is confident in its prediction for that class.

4. Which of the following steps are necessary to load an image for prediction in Keras?

0.8 / 1 point

- ☒ Resize the image to match the input shape of the model

 **Correct**

Correct! Resizing the image ensures that it matches the input shape expected by the model.

- ☒ Convert the image to a numpy array

 **Correct**

Correct! Converting the image to a numpy array is necessary for Keras to process it.

- ☐ Compile the model again with new parameters

- ☐ Import the image processing library

- ☒ Normalize the pixel values

 **Correct**

Correct! Normalizing pixel values is a common preprocessing step before making predictions.

You didn't select all the correct answers

5. What is the primary purpose of dropout regularization in neural networks?

1 / 1 point

- ☐ To increase the number of filters in the convolutional layers
- ☒ To prevent overfitting by randomly setting some neurons to zero during training
- ☐ To reduce the spatial dimensions of the input volume
- ☐ To normalize the input of each layer to improve training speed

 **Correct**

Correct! Dropout regularization helps prevent overfitting by randomly setting some neurons to zero during training, which forces the network to learn more robust features.